

Don't freeze: Finetune encoders for better Self-Supervised HAR

Vitor Fortes Rey
DFKI and RPTU
Kaiserslautern, Germany

Dominique Nshimiyimana
DFKI and RPTU
Kaiserslautern, Germany

Paul Lukowicz
DFKI and RPTU
Kaiserslautern, Germany

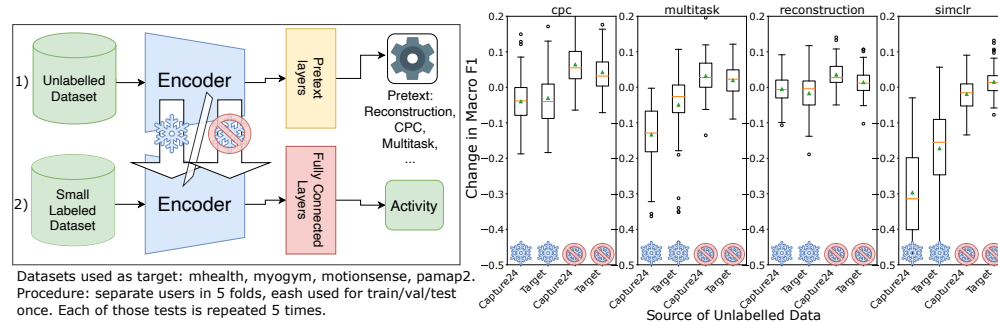


Figure 1: Change in performance given approach for using self-supervised learned representations in different methods.

ABSTRACT

Recently self-supervised learning (SSL) has been proposed in the field of human activity recognition (HAR) as a solution to the labelled data availability problem. The idea being that by using pretext tasks such as reconstruction or contrastive predictive coding, useful representations can be learned that then can be used for classification. Those approaches follow the pretrain, freeze and fine-tune procedure. In this work we investigate how a simple change - not freezing the representation - leads to substantial performance gains across pretext tasks. The improvement was found in all four investigated datasets and across all four pretext tasks and is inversely proportional to amount of labelled data. Moreover the effect is present whether the pretext task is carried on the Capture24 dataset or directly in unlabelled data of the target dataset.

CCS CONCEPTS

• Computer systems organization → Embedded systems; • Computing methodologies → Unsupervised learning.

KEYWORDS

self-supervised learning, human activity recognition

ACM Reference Format:

Vitor Fortes Rey, Dominique Nshimiyimana, and Paul Lukowicz. 2023. Don't freeze: Finetune encoders for better Self-Supervised HAR. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3594739.3610790>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0200-6/23/10.

<https://doi.org/10.1145/3594739.3610790>

1 RESULTS

We used four popular SSL methods [1–4] and compare the standard approach of fine-tuning with a frozen encoder versus fine-tuning the encoder and classification layers together, see Figure 1. We investigate learning representations in a bigger unlabelled HAR dataset (Capture24) versus in the target dataset. We found that compared to the baseline (no pre-training) using frozen representations helps in some cases, but those improvements are overshadowed by training the representation together with the classification layers (either on Capture24 or on the target dataset). Performance improves across all datasets for most pretexts (Figure 1), with the best results being with the cpc and Multi-task methods, even if their baseline results are not the best for that dataset. The exception is the SimCLR pretext, which decreases performance in the myogym and motionsense datasets when pretexts are done using Capture24 data. Interestingly, using a bigger dataset (Capture24) for SSL provided better results, probably due to the hyperparameter search for pretexts being performed over that dataset, but it is also possible that it being a bigger and more varied dataset makes up for the change in domain if the appropriate sampling is performed. We also investigated the effect of the number of epochs used when training the pretext task and found that Multi-task, reconstruction and simclr are less affected by the stopping point, with similar performance at the last epoch, while cpc can overfit, with better results at the epoch with best validation loss. It is also important to see the effect of labelled data availability, as we are interested in reducing labelled data requirements. We found that across datasets and pretexts we can often reach or surpass the performance of a baseline model with 50 or 100 windows per class by simply fine-tuning with 10 or fewer windows per class. This varies across pretexts and datasets, with cpc or Multi-task providing the most improvement.

ACKNOWLEDGMENTS

Research supported by grant H2020-EIC-FETPROACT-2019 #952091.

REFERENCES

- [1] Harish Haresamudram, David V. Anderson, and Thomas Plötz. 2019. On the Role of Features in Human Activity Recognition. In *Proceedings of the 2019 ACM International Symposium on Wearable Computers* (London, United Kingdom) (ISWC '19). Association for Computing Machinery, New York, NY, USA, 78–88. <https://doi.org/10.1145/3341163.3347727>
- [2] Harish Haresamudram, Irfan Essa, and Thomas Plötz. 2021. Contrastive Predictive Coding for Human Activity Recognition. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 2, Article 65 (jun 2021), 26 pages. <https://doi.org/10.1145/3463506>
- [3] Aaqib Saeed, Tanir Ozcelebi, and Johan Lukkien. 2019. Multi-task Self-Supervised Learning for Human Activity Detection. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 2 (jun 2019), 1–30. <https://doi.org/10.1145/3328932>
- [4] Chi Ian Tang, Ignacio Perez-Pozuelo, Dimitris Spathis, and Cecilia Mascolo. 2021. Exploring Contrastive Learning in Human Activity Recognition for Healthcare. arXiv:2011.11542 [cs.LG]